

Task 2 — Attack Taxonomy / Threat Model for O-RAN Fronthaul S-Plane

Five attack families targeting PTP timing distribution in 5G Open-RAN networks

Context & Scope

Your classifier currently discriminates two attack classes from TIMESAFE public data: **spoofing** (forged Announce messages) and **replay** (retransmitted Sync/Follow_Up). This taxonomy expands to five families, covering the full attack surface on PTP fronthaul timing. Two are proven; three are open — one of which (time-source manipulation) directly stresses your project's core novelty claim of distinguishing fault from attack.

1. Spoofing / Grandmaster Impersonation (Announce-based)

Mechanism: Attacker injects forged PTP Announce messages claiming superior clock attributes (lower priority1, better clockClass/clockAccuracy/offsetScaledLogVariance, lower priority2). The Best Master Clock Algorithm (BMCA) re-evaluates all available grandmasters and, if forged values win the comparison, every downstream clock re-parents to the attacker's false reference.

What changes in the signal: A new grandmasterIdentity appears in the traffic; stepsRemoved field shifts; parent and grandmaster clock-quality fields no longer reflect a trustworthy reference even though the protocol appears nominally healthy at the message level.

Detection clue: Grandmaster identity change not accompanied by a legitimate topology event (cable pull, planned failover); implausible jump in advertised clockClass/priority1 from a source with no prior track record; Announce messages arriving from a MAC/port that has never sourced Announce traffic before.

Status: ■ **Covered** — this is your existing 'spoo' class. References: TIMESAFE (ACM ToPS 2025); O-RAN.WG11.Threat-Model.O-R003.

2. Replay (Sync / Follow_Up Retransmission)

Mechanism: Attacker captures legitimate Sync and Follow_Up messages and retransmits them out of their original timing context. Because a slave's offset calculation depends on the *freshness* of the timestamps embedded in these messages relative to when they arrive, replayed messages corrupt the drift/offset/delay computation without needing to forge any new BMCA-winning content. The attack is pure message reuse, not forgery.

What changes in the signal: SequenceID reuse or out-of-order SequenceID stream; a Sync/Follow_Up pair whose correlation no longer matches a single logical exchange; timestamps that imply a network delay inconsistent with the real path characteristics.

Detection clue: linuxptp's PortServiceStats structure natively exposes sync_mismatch and followup_mismatch counters specifically designed for detecting this condition. No custom pcap parsing required — these counters tick whenever a Sync/Follow_Up correlation fails.

Status: ■ **Covered** — this is your existing 'sync'/replay class. References: TIMESAFE §Replay Attack; IEEE Std 1588-2019 §Sync/Follow_Up correlation.

3. Delay / MITM (Asymmetric Delay Injection)

Mechanism: A man-in-the-middle on the network introduces asymmetric delay on Sync and/or Delay_Req/Delay_Resp messages without altering their content. PTP's offset calculation assumes a near-symmetric path delay; injected asymmetry directly biases the computed offset. This is one of the hardest classes to catch because nothing in the message *content* is forged — the attack lives purely in the transmission delay.

What changes in the signal: Path delay (meanPathDelay) drifts away from its historical baseline or from a redundant-path estimate, without any corresponding physical or topology change. The slave observes a skewed round-trip time that doesn't match reality.

Detection clue: The academic approach is cyclic path-asymmetry analysis over redundant paths (PTPsec, Finkenzeller et al., arXiv:2401.10664): if you have more than one path between slave and master, comparing round-trip delay across paths reveals non-zero asymmetry that single-path monitoring cannot. A simpler, lower-fidelity signal is statistical deviation of meanPathDelay from a rolling baseline.

Status: ■ **Open** — not currently covered. This is the attack type being actively investigated by 2024–2025 research literature precisely because it evades content-based detection. References: Finkenzeller et al., PTPsec (arXiv:2401.10664); O-RAN Fronthaul Security using MACsec.

4. Denial of Service (Message Flooding on Master/Slave)

Mechanism: Attacker floods the master or a slave with excessive PTP traffic (Announce, Sync at abnormal rates far outside the configured logSyncInterval/logAnnounceInterval). Goal is to degrade or exhaust the clock's ability to serve legitimate timing, as distinct from impersonating it. This is a resource-exhaustion attack on the timing infrastructure.

What changes in the signal: Message-rate anomalies well outside configured cadence; PortServiceStats timeout counters (sync_timeout, announce_timeout, master_sync_timeout, master_announce_timeout) climbing steadily. The slave may begin timing out waiting for expected messages.

Detection clue: Rate-based anomaly detection is comparatively cheap: your existing 0.4 s windowing and 10-feature extraction pipeline could absorb a message-rate feature with modest engineering effort, since it doesn't require new parsing logic — just a counter over the window.

Status: ■ **Open** — not currently covered, and explicitly flagged by TIMESAFE as 'beyond PTP itself.' This makes it the safest novel-attack candidate: you're filling a gap they acknowledged, not contradicting their work. References: Open RAN Security: Challenges and Opportunities (arXiv:2212.01510); O-RAN Fronthaul Security using MACsec.

5. Time-Source Manipulation (Upstream GNSS Spoofing/Jamming or Compromised Grandmaster)

Mechanism: Rather than attacking the PTP message exchange itself, the attacker corrupts the reference the grandmaster trusts — most commonly by spoofing or jamming its GNSS receiver, or by compromising the grandmaster host directly. The PTP protocol then faithfully propagates a *wrong but internally consistent* time downstream. Nothing in the Sync/Announce exchange looks anomalous because the grandmaster genuinely believes its own input is trustworthy.

What changes in the signal: Not the PTP messages at all, at least not initially — the change is upstream at the GNSS receiver / grandmaster boundary. This is exactly why it's dangerous: it's invisible to a classifier that only watches PTP wire traffic. The protocol continues to look healthy.

Detection clue: GNSS receiver self-diagnostics is the honest answer. The O-RAN M-plane synchronization YANG model defines a gnss-sync-status leaf with values SYNCHRONIZED, ACQUIRING-SYNC, ANTENNA-DISCONNECTED, BOOTING, and ANTENNA-SHORT-CIRCUIT — the last two are physical-tamper indicators, not generic 'signal lost' states, designed for exactly this attack class. A secondary clue is holdover-oscillator drift: if the grandmaster free-runs and observed drift doesn't match known oscillator specs, that's suspicious.

Status: ■ **Open** — stresses H0/H1 novelty directly — Your simulator models 'GNSS loss/holdover' as a benign fault (H0). But a malicious GNSS spoof looks identical to a benign outage: both present as the GM losing its reference. Only upstream signal (antenna/receiver diagnostics) tells them apart. This is not just 'one more attack to bolt on' — it's a direct stress test of your paper's central claim. Recommend prioritizing this over #3/#4 for strategic value. References: O-RAN M-plane sync YANG (o-ran-sync.yang); TEXBAT (UT Austin) GNSS-spoofing dataset; Humphreys et al., Texas Spoofing Test Battery (ION GNSS 2012).

Summary Table — All Five Attacks

#	Attack	Signal Effect	Best Detection Clue	Status
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1	Spoofing (Announce/BMCA)	New GM identity/quality fields	Unexplained grandmaster change without topology event	■
2	Replay (Sync/Follow_Up)	SequenceID reuse; stale correlation	sync_mismatch / followup_mismatch counters (linuxptp native)	■
3	Delay / MITM	meanPathDelay drifts from baseline	Cyclic path-asymmetry analysis; redundant-path comparison	■
4	DoS (master/slave flood)	Message-rate anomaly; timeout counters climb	Message-rate feature; PortServiceStats timeout fields	■
5	Time-source manipulation	No PTP anomaly (upstream GNSS/GM only)	gnss-sync-status fields (ANTE NNA-DISCONNECTED, etc.); holdover drift	■

Priority Ranking for Novel-Attack Fix

Rank	Attack	Strategic Value	Engineering Cost	Rationale
1	Time-source manipulation (#5)	Highest	Moderate	Directly exercises your H0/H1 novelty; needs gnss-sync-status feature channel (simulator doesn't have this today)
2	DoS/flooding (#4)	Medium	Lowest	Rate feature fits your existing 0.4 s windowing with no new parsing; TIMESAFE explicitly leaves it open so no overlap risk
3	Delay/MITM (#3)	Highest (research)	Highest	Engages 2024–2025 literature directly; requires redundant-path topology and cyclic-asymmetry logic you may not have on single-machine testbed

Verification Checklist

- Confirm current O-RAN WG11 threat model version at <https://specifications.o-ran.org> (latest supersedes O-R003-v06.00 from June 2023)
- If pursuing #5 (time-source manipulation) with real data: decide whether TEXTBAT GNSS-SDR integration (requires RF capture) is worth the effort vs. adding synthetic gnss-sync-status feature to simulator
- Cross-reference attack mitigation strategies against your digital-twin recovery actions (switchover, GPS failover, holdover, rogue-master isolation, rerouting)